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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

- ☐ Gradient Boosting (unit? unit=95&lesson=96)
- ☐ Random Forests I (unit? unit=95&lesson=97)
- ☐ Naive Bayes (unit? unit=95&lesson=98)
- ☐ Bayesian Networks (unit?)

Week 8: Assignment 8 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) In AdaBoost, we re-weight points giving points misclassified in previous iterations more weight. **1 point**
 Suppose we introduced a limit or cap on the weight that any point can take (for example, say we introduce a restriction that prevents any point's weight from exceeding a value of 10). Which among the following would be an effect of such a modification? (Multiple options may be correct)

- ☐ We may observe the performance of the classifier reduce as the number of stages increase
- ☐ It makes the final classifier robust to outliers
- ☐ It may result in lower overall performance

2) A and B are Boolean random variables. **1 point**

Given:

$P(A = \text{True}) = 0.3, P(A = \text{False}) = 0.7, P(B = \text{True}|A = \text{True}) = 0.4, P(B = \text{False}|A = \text{True}) = 0.6, P(B = \text{True}|A = \text{False}) = 0.6, P(B = \text{False}|A = \text{False}) = 0.4$. Calculate $P(A = \text{True}|B = \text{False})$ by Bayes rule.

- ☐ 0.49
- ☐ 0.39
- ☐ 0.37
- ☐ 0.28
- ☐ None of the above

3) Which of the following method(s) is not inherently sequential? **1 point**

- ☐ Gradient Boosting
- ☐ Committee machines
- ☐ AdaBoost

4) In a tournament classifier with N classes. What is the complexity of the number of classifiers we require? **1 point**

- ☐ $\mathcal{O}(N)$
- ☐ $\mathcal{O}(N^2)$
- ☐ ...

unit=95&lesson=99)

☐ Multiclass Classification (unit? unit=95&lesson=100)

☐ **Practice: Week 8: Assignment 8 (Non Graded) (assessment? name=271)**

☒ Quiz: Week 8: Assignment 8 (assessment? name=300)

☐ Week 8 Feedback Form : Introduction To Machine Learning (unit? unit=95&lesson=289)

☒ Week 8: Solution (unit? unit=95&lesson=227)

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

Text Transcripts ()

Download Videos ()

Books ()

Problem Solving Session - July 2024 ()

$\mathcal{O}(N \cdot \log(N))$

☐ $\mathcal{O}(\log(N))$

Check Answers and Submit

